Operation Management

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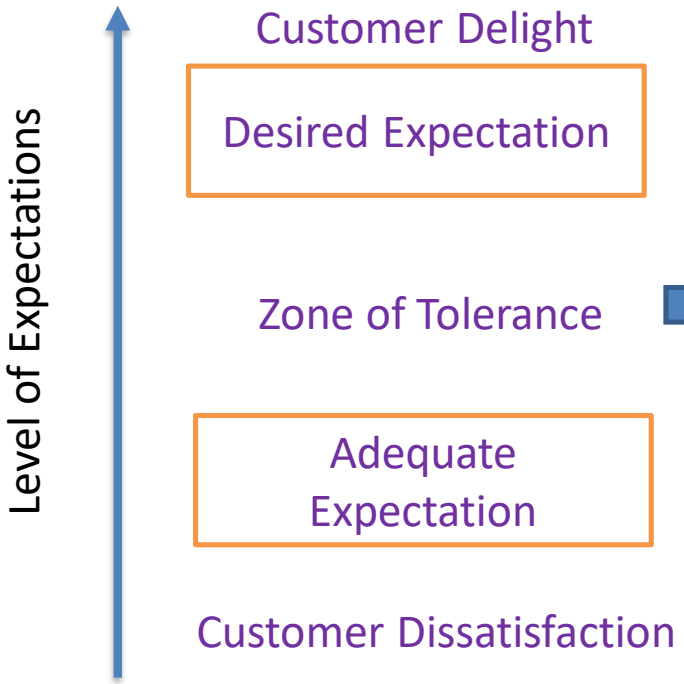
<MBAZG526/QMZG526, Operation Management>
Lecture No. 13

Quality

- The ability of a product or service to meet customer expectations.



- The extent to which customers recognize and are willing to accept.
- Different customers have different zone of tolerance for same product/services.
- Same Customer have different zone of tolerance for different product/services.

Quality Characteristics

- Performance
- Aesthetics
- Special features
- Conformance
- Reliability
- Durability
- Perceived quality
- Serviceability
- Consistency

Quality Gurus

Dr. Walter A. Shewart.

- Scientist at Western Electric and Bell Telephone Lab.
- Developed control chart theory with control limit, assignable and chance cause of variations.
- Developed PDSA cycle (Plan-Do-Study-Act).
- Authored famous book " Economic control of Quality of Manufactured product"

Quality Gurus

Dr. W. Edward Deming.

- Made a significant contribution to Japan's reputation for innovative product of high quality.
- Provides Deming 14 principles of effective management.
- Authored famous books " Out of Crisis" and "Productivity and competitive position."

Deming's 14 principles

TABLE 6.2 Deming's 14 Points for Implementing Quality Improvement

1. Create consistency of purpose.
2. Lead to promote change.
3. Build quality into the product; stop depending on inspections to catch problems.
4. Build long-term relationships based on performance instead of awarding business on the basis of price.
5. Continuously improve product, quality, and service.
6. Start training.
7. Emphasize leadership.
8. Drive out fear.
9. Break down barriers between departments.
10. Stop haranguing workers.
11. Support, help, and improve.
12. Remove barriers to pride in work.
13. Institute a vigorous program of education and self-improvement.
14. Put everybody in the company to work on the transformation.

Source: Deming, W. Edwards. *Out of the Crisis*, pp. 23–24, © 2000 W. Edwards Deming Institute, published by The MIT Press. Reprinted by permission.

Source: Heizer, Render, Muckson

Quality Gurus

Dr. Joseph M Juran

- Introduces Juran Trilogy—Quality Planning, Quality Control and Quality Improvement.
- He recommended that the investment on any project should be based on ROI.
- He published Juran's Quality control handbook.

Quality Gurus

Dr. Armand Feigenbaum

- Total Quality control is necessary to achieve productivity, market penetration and competitive advantage.
- He recommended that quality begins by identifying customer's requirements and ends with a product or service in the hands of satisfied customer.
- He authored Total Quality Control—which states Quality Leadership, Quality technologies and organizational commitment.

Quality Gurus

Dr. Phillip Crossby

- Originates the concept of zero defect which includes 14 steps to quality improvement.
- He believe that " doing it right the first time."
- Introduces "Absolute of Quality management", which states that Quality is conformance to requirements, which comes from prevention of errors. The quality performance standard is zero defect.

Quality Gurus

Dr. Kaouru Ishikawa

- Developed the concept of Quality circle in Japan.
- Developed C&E diagram/ Ishikawa diagram/ Fish bone diagram.
- Introduces Cross functional management.

Quality Gurus

Dr. Genichi Taguchi

- Developed Taguchi methods of DOE.
- Developed Signal to Noise Ratio.
- Developed Loss function that combines cost, target and variation.
- Making processes/products insensitive to environmental factors.

Malcolm Baldrige National Quality Award

- Awarded in Six Categories—Manufacturing, Service, Small business, education, healthcare and non-profit.
- Winners include such firms as
- Motorola, Milliken, Xerox, FedEx, Ritz-Carlton Hotels, AT&T, Cadillac, and Texas Instruments.

Cost of Quality (COQ)

COQ = Prevention Cost + Appraisal Cost + Internal Failure Cost + External Failure Cost

- **Prevention cost:** Everything the company spends to prevent errors, (e.g. customer survey, design review, supplier rating, process control, quality system audit, training).
- **Appraisal cost:** Everything the company spends to check that the product/service meets the quality requirements (e.g. inspection and testing, product quality audit, review of test and inspection data)

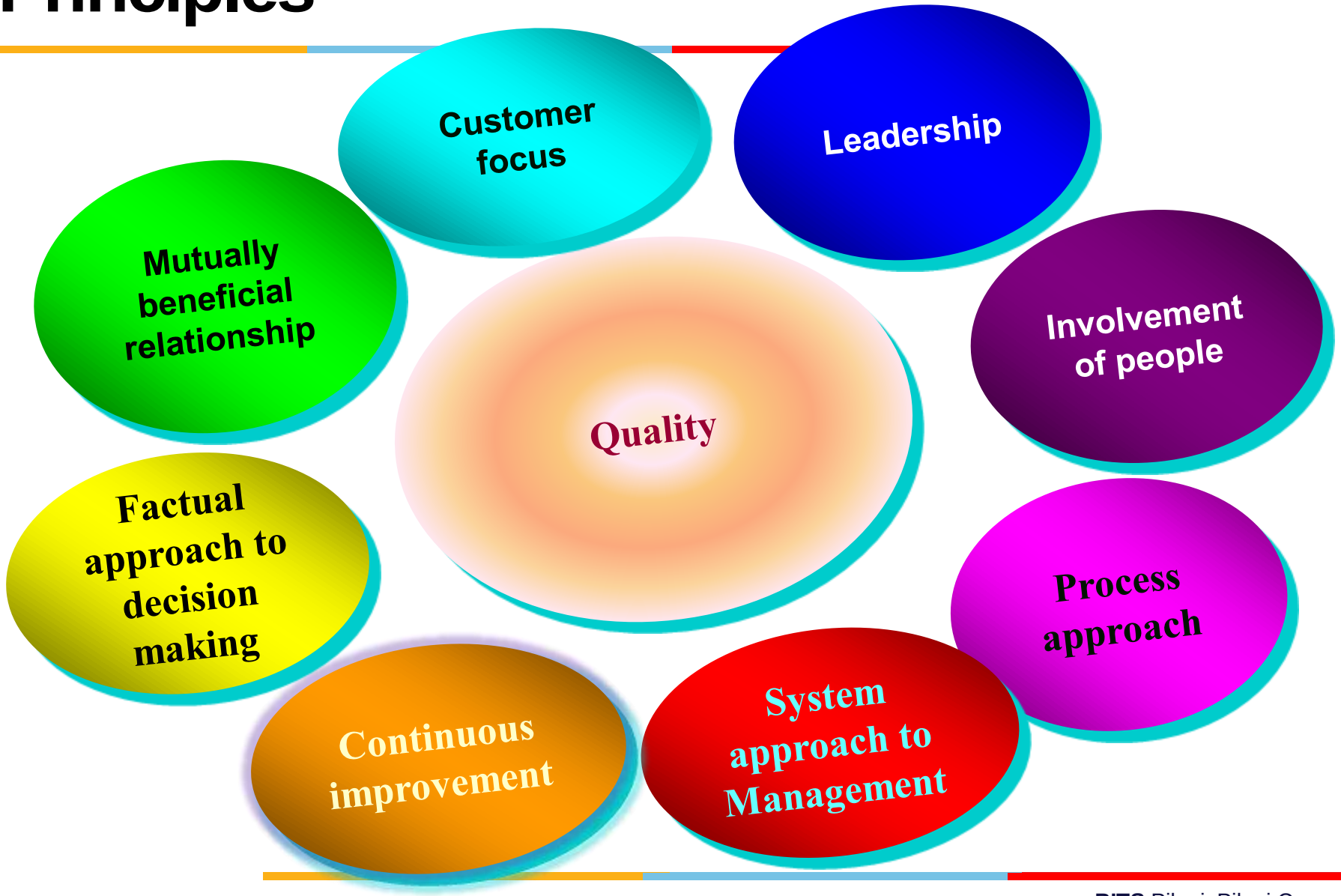
Cost of Quality (COQ)

- **Internal failure cost:** Cost of in-house failures (e.g. scrap, downgrading, overtime, process reset)
- **External failure cost:** Cost of failures during transit and at customer end (e.g. warranty, product recall, transit damage/shortage, handling customer complaints)

Quality Verification & Validation

Feature	Verification	Validation
Focus	Internal aspects and Design aspects	Product specification and user needs
Timing	Before Validation	After Verification
Question	Are we building the product right	Are we building the right product
Objective	Prevent errors	Detect errors
Responsibility	Design and Operation Team	QA Team

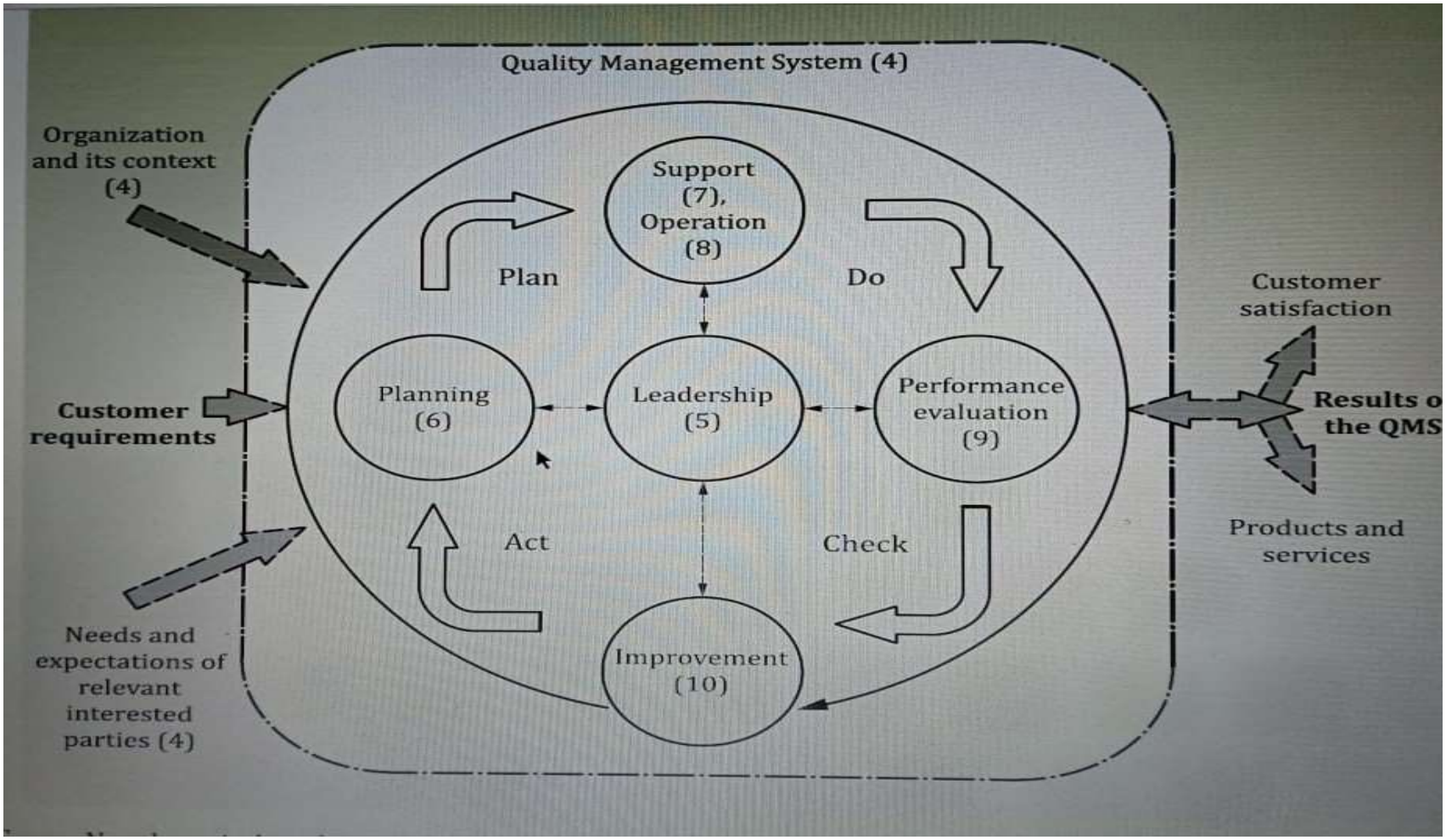
Quality Management Principles



Total Quality Management

- Management of an entire organization so that it excels in all aspects of products and services that are important to the customer.
- Seven concepts for an effective TQM program:
- (1) Continuous improvement, (2) Six Sigma, (3) employee empowerment, (4) benchmarking, (5) just-in-time (JIT), (6) Taguchi concepts, and (7) knowledge of TQM tools.

Continuous Improvement



Six Sigma

Management Philosophy

- View process from customer's point of view.
- Continual improvement.
- Integration of production and quality.
- Satisfying customer need.

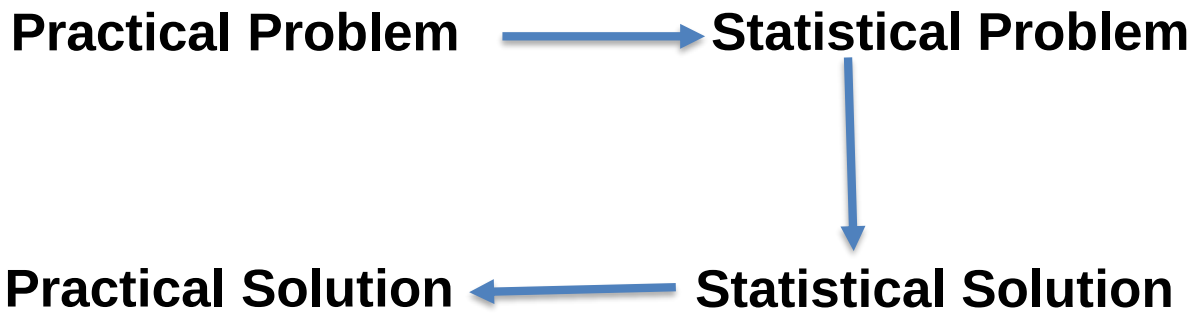
Metric

- 3.4 defects per million opportunities or 99.99996 % defect free production

Six Sigma

Set of Tools

- Statistical Process Control (SPC).
- Control Chart.
- Failure Mode of Effect Analysis (FMEA).
- Process mapping.



Six Sigma : When Data Exist

D = Define the problem and the objective.

M = Measure the extent of the problem.

A = Analyze the process and define the factors of influence.

I = Improve the process.

C= Control the improved process.

Six Sigma : When Data Do Not Exist

D = Define the problem and the objective.

M = Measure the extent of the problem.

A = Analyze the process and define the factors of influence.

D = Design the process.

V= Verify the designed process

Employee Empowerment

- Enlarging employee jobs so that the added responsibility and authority is moved to the lowest level possible in the organization.
- Techniques for building employee empowerment include (1) building communication networks that include employees; (2) developing open, supportive supervisors; (3) moving responsibility from both managers and staff to production employees; (4) building high morale organizations; and (5) creating such formal organization structures as teams and quality circles.

Quality Circles

- A group of employees meeting regularly with a facilitator to solve work-related problems in their work area.
- To give opportunity to employees to use their wisdom and creativity.
- To reduce error and enhance quality, job improvement and productivity.

Benchmarking

- A group of employees meeting regularly with a facilitator to solve work-related problems in their work area.
- Benchmarking is the systematic search for best practices, innovative ideas, and highly effective procedures.

Types of Benchmarking

Process Benchmarking : Comparison of own processes with leading service providers.

Ex: Florida Power and Lights in pursuit of Deming award benchmarked Kansai Electric of Japan.

Internal Benchmarking: Within organization.

Ex: Multisite operation processes, comparison with different machines.

Generic Benchmarking: Comparison of outstanding processes irrespective of industry.

Ex: Restaurant chain kitchen benchmarking kitchen design of Submarine.

Competitive Benchmarking: Specific competitor to competitor comparison for a product, service or function of interest.

Ex: Ford benchmarked against Mazda and ended up removing its NVA

Just In Time

- Just in Time (JIT) manufacturing is a philosophy which eliminates waste associated with time, labor, and storage space.
- The company produces only what the customer requests, to actual orders, not to forecast.
- Raw materials are received just in time to go into production, manufacturing parts are completed just in time to be assembled into products, and products are completed just in time to be shipped to customers.

Taguchi Concepts

- Robust Quality : Products that are consistently built to meet customer needs despite adverse conditions in the production process.
- Quality loss function (QLF) : A mathematical function that identifies all costs connected with poor quality and shows how these costs increase as output moves away from the target value.

Taguchi Concepts

- Orthogonal array design that work on Signal to Noise Ratio.
- It has three subsets:
- Nominal is Best.
- Larger the Better.
- Smaller the Better.

Knowledge of TQM Tools

- 7 QC tools

1. TALLY/CHECK SHEET
2. HISTOGRAM
3. PARETO DIAGRAM
4. CAUSE AND EFFECT DIAGRAM
5. SCATTER DIAGRAM
6. FLOW CHART
7. CONTROL CHART

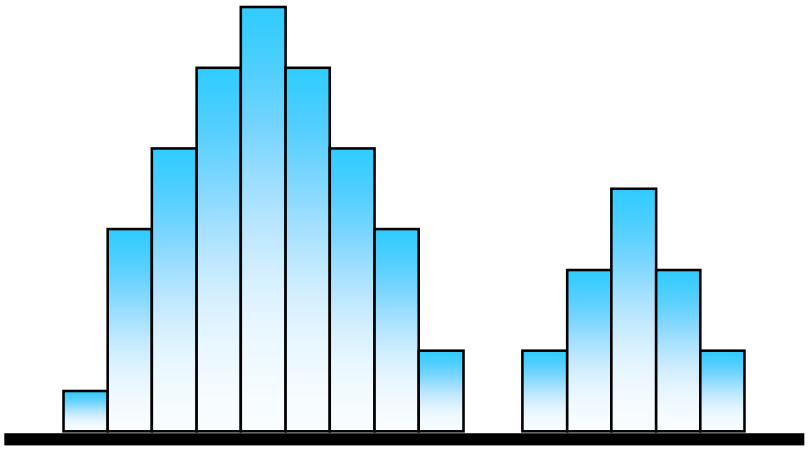
Check Sheet

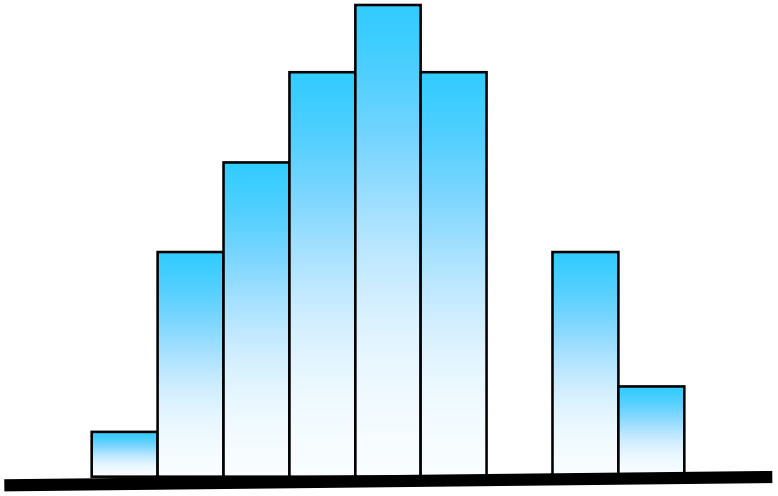
(a) *Check Sheet*: An organized method of recording data

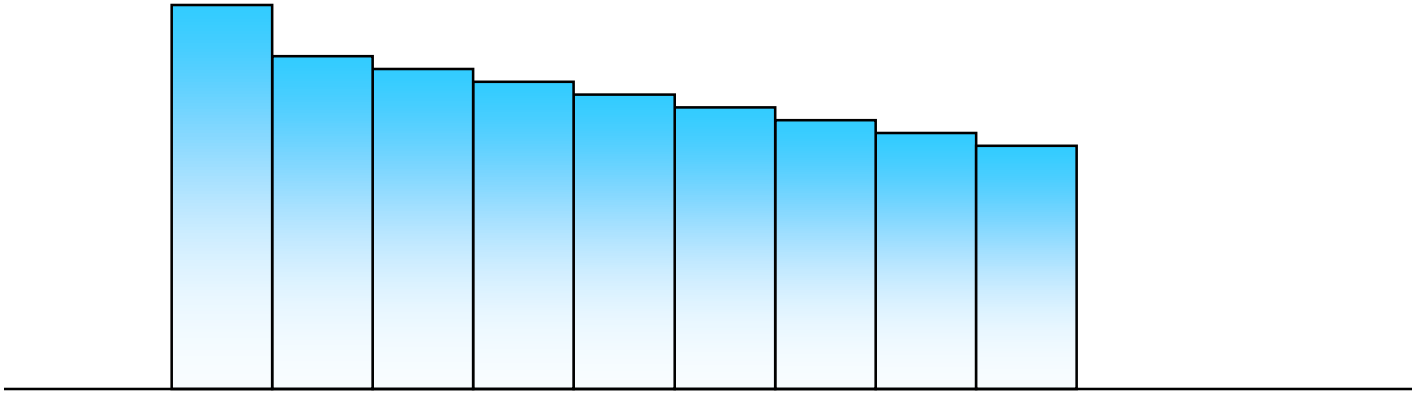
	Hour							
Defect	1	2	3	4	5	6	7	8
A	///	/		/	/	/	///	/
B	//	/	/	/			//	///
C	/	//					//	////

Source: Heizer, Render, Muckson

Histogram : Your Inference







Pareto Diagram

- The Pareto diagram is named after Vilfredo Pareto, a 19th century Italian economist who postulated that a large share of wealth is owned by a small percentage of the population.
- This basic principle translates well into quality problems– most quality problems result from a small number of causes.
- Quality experts often refer to the principle as the 80/20 rule; that is 80% of the problems are caused by 20% of the potential sources.

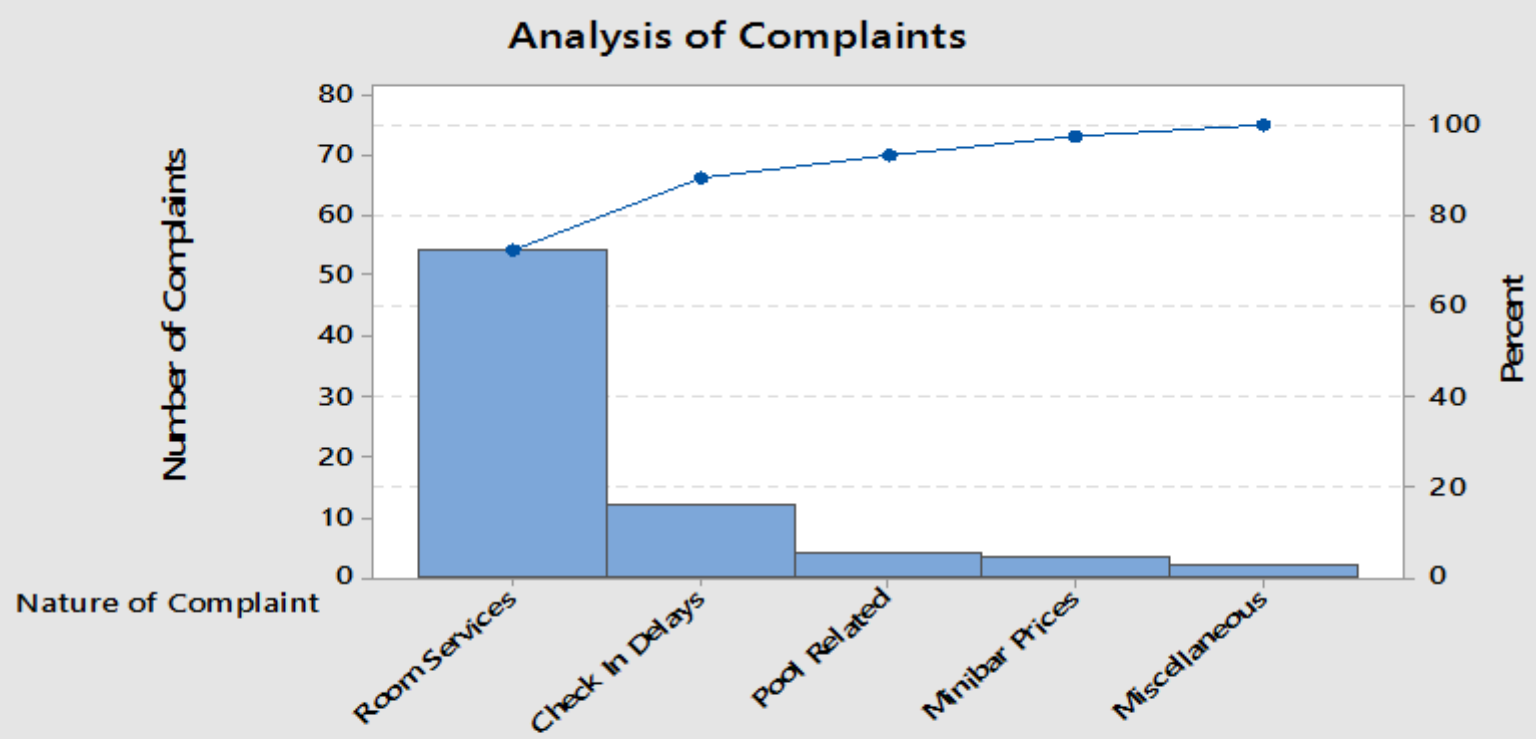
Pareto Diagram

- A Pareto diagram puts data in a hierarchical order which allows the most significant problems to be corrected first.
- The Pareto analysis technique is used primarily to identify and evaluate nonconformities, although it can summarize all types of data.
- It is perhaps the diagram most often used in management presentations.

Class Exercise

- The Hard Rock Hotel in Bali has just collected the data from 75 complaint calls to the general manager during the month of October. The manager wants to prepare an analysis of the complaints.
- The data provided are room service, 54; check-in delays, 12; hours the pool is open, 4; minibar prices, 3; and miscellaneous, 2.
- Which are the 20 % causes causing 80 % of defect?

Pareto Diagram

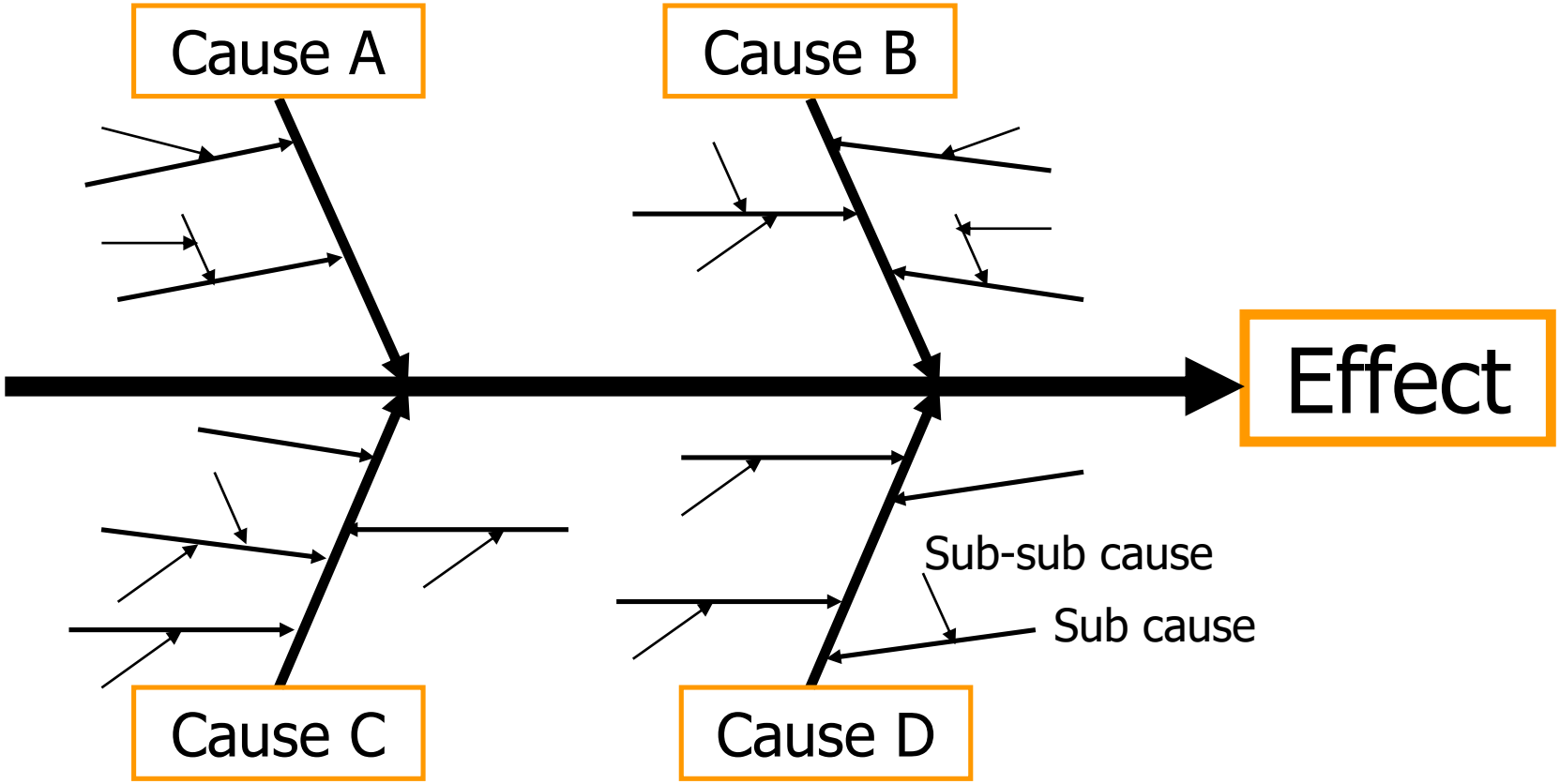


Nature of Complaint	Room Services	Check In Delays	Pool Related	Minibar Prices	Miscellaneous
Number of Complaints	54	12	4	3	2
Percent	72.0	16.0	5.3	4.0	2.7
Cum %	72.0	88.0	93.3	97.3	100.0

Cause and Effect Diagram

- Manufacturing
  - 6M (Man, Machine, Material, Method, Measurement and Mother Earth)
- Service
  - 4P (Policy, Procedure, People, Planet).

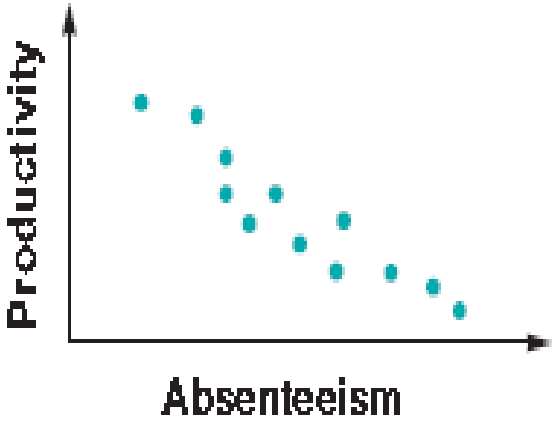
Cause and Effect Diagram



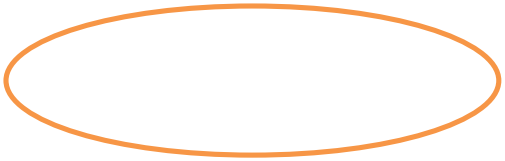
Scatter Diagram

- The scatter diagram shows the pattern of relationship between two variables that are thought to be related.
- The closer the points hug diagonal line the more closely there is a one to one relationship.

(b) *Scatter Diagram:* A graph of the value of one variable vs. another variable

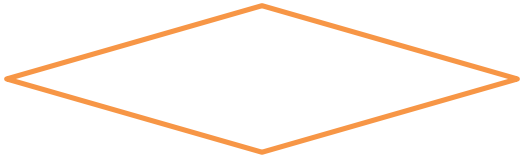


Flow Chart



Start Point & End Point

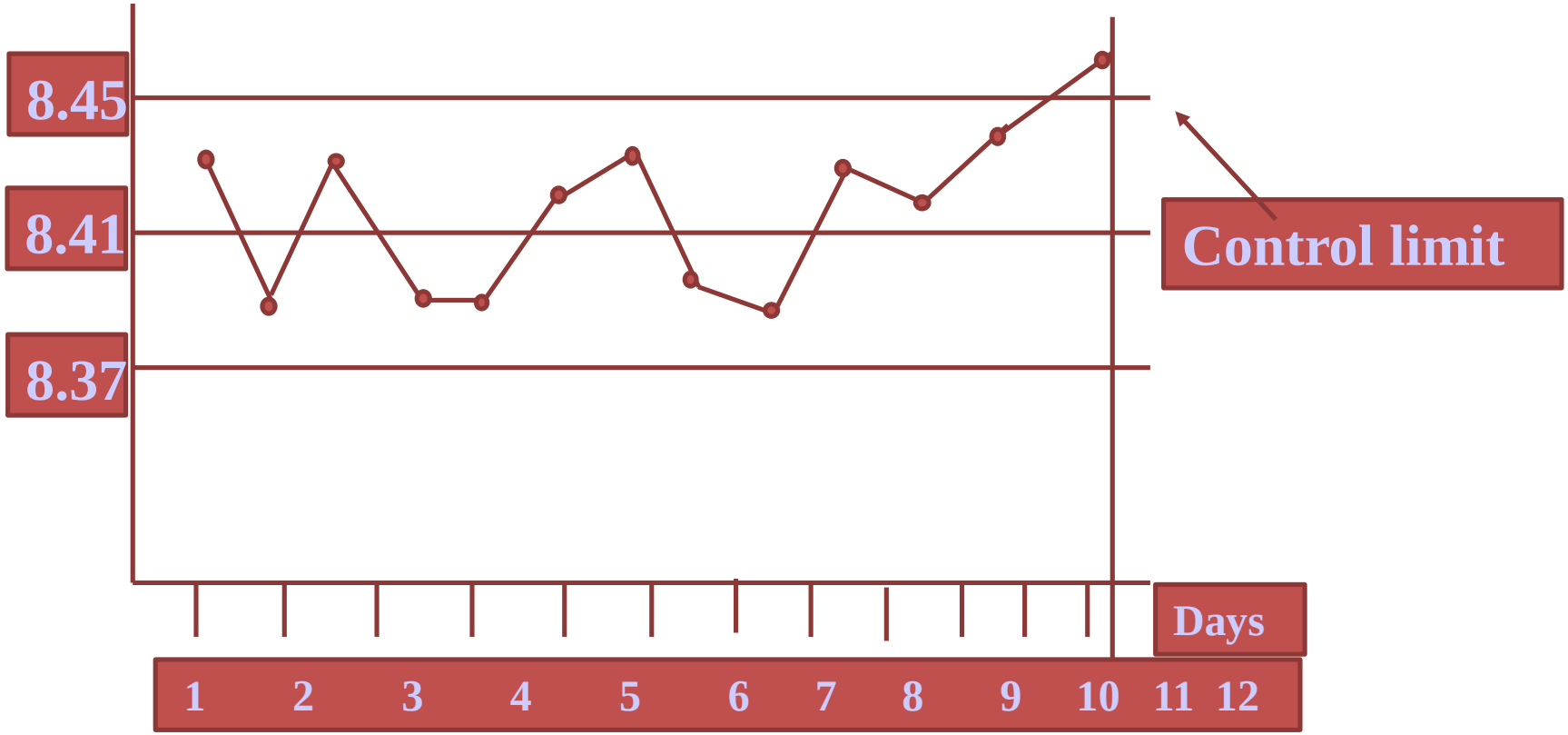
Sequence of Steps



Decision Points

Direction

Control Chart



Any Question

Quality Control Techniques

1. Inspection
2. Statistical Process Control

Inspection

- A means of ensuring that an operation is producing at the quality level expected.
- Attribute inspection : An inspection that classifies items as being either good or defective.
- Variable inspection : Classifications of inspected items as falling on a continuum scale, such as dimension or strength.

Source Inspection

- Also known as source control
- The next step in the process is your customer
- Ensure perfect product to your customer

Source Inspection

- **Poka-yoke** is the concept of foolproof devices or techniques designed to pass only acceptable products
- **Checklists** ensure consistency and completeness

Statistical Process Control

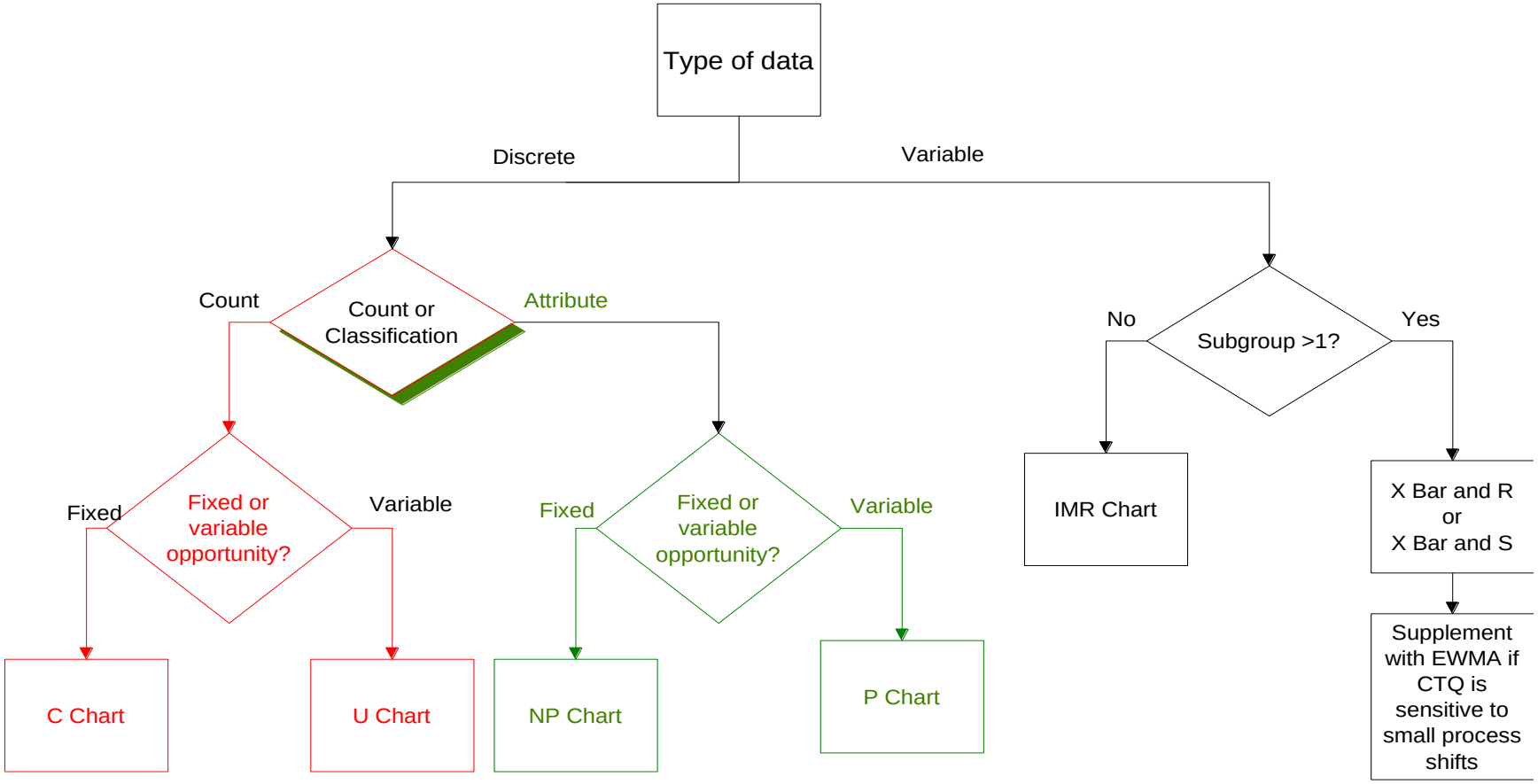
- A process used to monitor standards by taking measurements and corrective action as a product or service is being produced.
- Any deviation from the process or product or customer requirements are termed as Variations.
- Variations are of two types : Common cause of variations and Special/ Assignable Causes of Variation.
- Common cause of variations are natural.
- Special cause of variations are due to assignable causes.

Central Limit Theorem

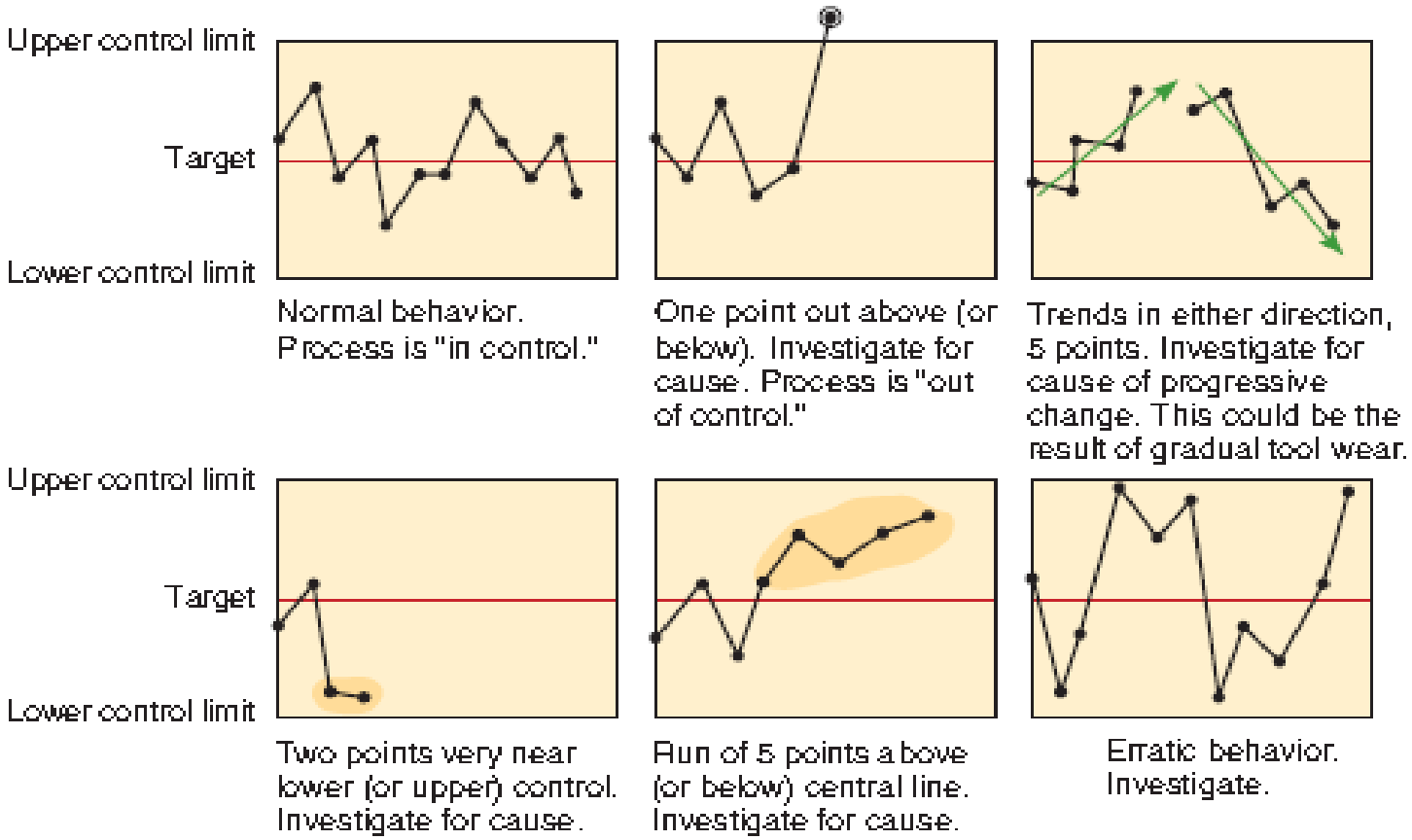
- The central limit theorem (CLT) states that, regardless of the shape of the population, the sampling distribution of the mean is approximately normal if the sample size is sufficiently large.
- The approximation improves as the sample size gets larger.

Control Chart

- A graphical presentation of process data over time.



Control Chart



Process Capability

- Process Capability is a simple tool which helps us to determine if a process is capable of meeting customer requirement or not.
- It helps us to determine the percent of products/ services not meeting the customer requirement.
- To measure process capability process should be in control.

Process Capability Ratio

$$Cp = (USL - LSL)/\ 6\sigma$$

$$Cp = \frac{Tolerance}{Natural\ Process\ Variation}$$

Process Capability (Index)

$$Cpk = \min [(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$$

Sigma Level = 3*Cpk or 3*Cp

Cp= Potential Process Capability

Cpk= Actual Process Capability

Actions For Non-Conformity

- CAPA : Pareto Chart, QMS
- Root Cause Analysis : 5 Whys, Fishbone Diagram, etc.
- Continuous Improvement : Process Capability, Control Chart, SPC, Lean Six Sigma.

Any Question

TQM In Services

- Service quality is more difficult to measure than the quality of goods
- Service quality perceptions depend on
  1) *Intangible differences between products*
  2) *Intangible expectations customers have of those products*

Service Quality

The operations manager must recognize:

- The tangible component of services is important
- The service process is important
- The service is judged against the customer's expectations
- Exceptions will occur

Determinants of Service Quality

Table 6.5
Reliability involves consistency of performance and dependability
Responsiveness concerns the willingness or readiness of employees to provide service
Competence means possession of the required skills and knowledge to perform the service
Access involves approachability and ease of contact
Courtesy involves politeness, respect, consideration, and friendliness
Communication means keeping customers informed and listening to them
Credibility involves trustworthiness, believability, and honesty
Security is the freedom from danger, risk, or doubt
Understanding/knowing the customer involves making the effort to understand the customer's needs
Tangibles include the physical evidence of the service

TQM in Services

- **Service recovery** —Training and empowering frontline workers to solve a problem immediately.
- **SERVQUAL**—A popular measurement scale for service quality that compares service expectations with service performance.

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 6
2. Operation Management by William J Stevenson—Chapter 9, 10
3. Six Sigma Master Black Belt Handbook—ASQ.
4. Statistical Process Control by Douglas C. Montgomery.

THANK YOU